Program : Diploma in Communication and Computer Networking	
Course Code : **6329C**	Course Title: **IoT Fundamentals and Design Lab**
Semester: **6**	Credits: **1.5**
Course Category: **Engineering Sciences**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Setup a basic IoT hardware.
- Apply IoT concept in simple real life applications.
- Apply IoT concepts in advance applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Overview of computer networks and topologies	4322	Computer Networks	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Familiarization of basic IoT hardware.	4	Applying
CO2	Setup basic IoT hardware.	9	Applying
CO3	Apply the IoT concept in simple real life applications.	12	Applying
CO4	Apply IoT concepts in advanced applications.	12	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3		2		
CO2	3	3	3		2		
CO3	3		3		2		
CO4	3		3		2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Familiarization of basic IoT hardware.**		
M1.01	Familiarization of of Arduino board ,ESP8266 wifi module,LM35,L293D motor driver	4	Applying
CO2	**Setup a basic IoT hardware.**		
M2.01	**Connect Arduino board with internet.** Basic IoT setup with Arduino and ESP8266: Connection of Arduino board with ESP8266 wifi module, interfacing Arduino with ESP8266 using AT commands like UART, CWMODE, CWLAP, CWJAP, CIPMUX, CIPSERVER, CIFSR. Connecting Arduino to access-point with LAN/internet with static IP. Checking TCP connection with Arduino over LAN/internet.	3	Applying
M2.02	**Demonstrate the working of simple IoT task of LED control.** Writing first IoT based Program on Arduino: To control an LED connected to an Arduino: -Write a basic program (i.e. html code) in a PC for creating command buttons on a browser window. -Write and upload the Arduino code for ON/OFF control of LED. -Run the program of Arduino and give the browser based command to	3	

	control the LED.		
	Series Test– I	3	
CO3	**Apply IoT concept in simple real life applications.**		
M3.01	**Implement IoT based temperature logger.** Cloud based data logging: IoT based Temperature logger using ThingSpeak (Or any other cloud service) Arduino, LM35 and ESP8266 - Connection of LM35 with Arduino board (which is already connected to internet/intranet with the help of ESP8266) - Setting up cloud based account (Thingspeak etc.) or any other IoT cloud service/server. - Write and upload Arduino temperature data logger program using LM35, given IoT cloud service and ESP8266. - View and verify the temperature logs on the IoT cloud service.	6	Applying
M3.02	**Implement IoT based home automation system.** IoT based home automation - Connection of relays with Arduino board (which is already connected to internet/intranet with the help of ESP8266) - Writing cloud based or local executable code (i.e. plain html code) to communicate with the above Arduino board. - Execute the above code to send the ON/OFF control commands via internet/intranet to the relays connected to different pins of the Arduino board which ultimately will switch ON/OFF the electrical/electronic appliances.	6	Applying
CO4	**Apply IoT concepts in advance applications.**		

M4.01	**Implement IoT based street light control system** IoT Based Street Light Control - Connection of LDR and relays (connected to street lights) with Arduino board (which is already connected to internet/intranet with the help of ESP8266) - Writing cloud based or local executable code (i.e. plain html code) to communicate with the above Arduino board. - Execute the above code to sense the ambient light near the street light and if it is less/greater than the predefined threshold level then send the ON/OFF control commands via internet/intranet to the relays connected to different pins of the Arduino board which ultimately will switch ON/OFF the street lights.	6	Applying
M4.02	**Apply IoT concepts in speed control of DC motor.** Connection of L293D motor driver (connected to and DC motor) with Arduino board (which is already connected to internet/intranet with the help of ESP8266) - Writing cloud based or local executable code (i.e. plain html code) to communicate with the above Arduino board. - Executing the above code to send the instructions to the above Arduino board which in turn generates PWM signal to be fed to the motor driver and hence control the speed of DC motor.	6	Applying
	Series Test –II	4	
	Open Ended Experiments / micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.
Compulsory for CIA
The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text / Reference

T/R	Book Title/Author
R1	Internet of Things with Arduino Cookbook,Marco Schwartz,Packt Publishing Ltd.
R2	Internet of Things with Arduino Blueprints,Pradeeka Seneviratne,Packt Publishing Ltd.
R3	Internet of Things: A Hands On Approach,Arshdeep Bahga and Vijay Madisetti,Universities Press (India) Private Limited

Online Resources

Sl.No	Website Link
1	spoken-tutorial.org
2	nptel.ac.in
3	swayam.gov.in